

Using the data health systems already collect to make high-stakes care safer and more efficient.

### WHO WE ARE

**Afferent Health is an independent scientific and clinical R&D intelligence firm.** We enable health systems, federal partners, and technology teams to transform complex clinical data into deployable decision-support capabilities by integrating advanced ML/AI development with the specialized expertise and personnel required for execution.

We learn from the whole record - waveforms, monitor streams, laboratory results, longitudinal EHR data, and clinical text - where the data are richest and the stakes are highest. We earn trust through calibration, physiologic checks, subgroup and transfer testing, negative controls, disclosed limits, and human-use design.

### CORE CAPABILITIES

- **Complete-record clinical intelligence.** Cohort definition, record linkage, governance, and analysis across waveforms, monitor numerics, labs, EHR data, clinical text, and longitudinal history.
- **Multimodal waveform and signal processing.** Learning from raw arterial pressure, capnography, pulse oximetry, and EEG morphology - not only the summary numbers a monitor reports.
- **Accountable ML/AI and clinical TEVV.** Prototype development, calibration, subgroup and site-transfer testing, explainability, abstention, safety analysis, negative controls, drift monitoring, and test/evaluation/verification/validation support.
- **Perioperative, ICU, and acute-care intelligence.** Risk stratification and monitoring for surgical outcomes, biological age and reserve, delirium, deterioration, kidney injury, respiratory events, readmission, and care transitions.
- **Clinical workflow and decision-support design.** Turning clinical requirements into usable tools for transplant readiness, clinical summaries, release gates, surveillance, OR flow, coding review, and care navigation.
- **Research-to-product and value translation.** Evidence synthesis, data-feasibility review, study design, proof-of-concept planning, prospective validation, and auditable economic-impact and budget models.
- **Clinical solutioning for prime-led federal teams.** PWS-by-PWS clinical workflow, safety and validation, data assumptions, acceptance metrics, SME staffing, and the clinical portion of technical approaches and RFI/RFP responses.

### DIFFERENTIATORS

- **Physician-scientist depth.** Clinical validity, workflow fit, and measurable impact are designed in from the start by clinicians and scientists who understand both the signal and the decision it must support.
- **Existing data first.** When a health system already captures the signal, we make that record more useful before asking for new hardware or unnecessary workflow.

### PARTNER ECOSYSTEM



### COMPANY DATA

| LEGAL ENTITY                                          | PRIMARY NAICS                                                | ADDITIONAL NAICS                                                                                                                                                         | CONTACT                                          |
|-------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Afferent Health LLC<br>New York, NY<br>Small business | 541690<br>Other Scientific and Technical Consulting Services | 541715 - Research and Development in the Physical, Engineering, and Life Sciences (except Nanotechnology and Biotechnology)<br>541512 - Computer Systems Design Services | afferenthealth.com<br>contact@afferenthealth.com |

### KEY PERSONNEL AND ENGAGEMENT

- **D. K. Adrian Williams, MSc - Founder & Managing Principal.** MD-PhD candidate focused on computational neuroscience and signal processing, with work spanning multimodal physiology, explainable clinical AI/ML, pulse oximetry, capnography, EEG, research-to-product translation, and day-to-day operations.
- **Joseph R. Scarpa, MD-PhD - Medical Consultant.** Assistant professor of anesthesiology and NIA-funded principal investigator of an active R03 on biological age and surgical outcomes (R03AG095722), with expertise in perioperative physiology and clinical translation.
- **Ruben Coen-Cagli, PhD - Scientific & AI Consultant.** Associate professor of systems and computational biology and neuroscience; NIH/NEI-funded principal investigator and active R01 co-PI in computational neuroscience, probabilistic modeling, and representation learning (R01EY030578).
- **Delivery model.** Afferent leads clinical problem formulation, evidence, modeling, validation, acceptance metrics, and specialist inputs. Prime partners lead capture, pricing, contracts, engineering, integration, sustainment, and transition.

### CURRENT WORK AND PRODUCT DIRECTIONS

- **Surgical Intelligence.** Biological age and reserve, delirium prevention, perioperative risk, failure-to-rescue and readmission, ICU escalation, and OR readiness or flow.
- **Transplant decision support.** LLM-assisted referral readiness, packet quality, clinical navigation, and review support designed around documented transplant-service friction.
- **Clinical AI assurance.** TEVV and release gates for calibration, bias, missingness, drift, cross-site transfer, explanation, human review, and audit-ready evidence.
- **Physiologic intelligence.** Pulse-oximetry equity, capnography, arterial-waveform, EEG, and multimodal decompensation methods built from signals already present in routine care.
- **Clinical data and operations.** Clinical-summary quality, coding and appeals support, opioid-safety surveillance, workforce and rotation operations, and data-trust or reconciliation workflows.
- **Federal clinical R&D work packages.** Evidence-backed concepts and clinical or scientific contributions for pilots, task orders, grants, and R&D programs within prime-led capture and delivery teams.

*Afferent leads clinical problem formulation, evidence, modeling, validation, and SME inputs; prime partners lead capture, pricing, contracts, engineering, integration, and transition.*